

Structured summary: Three-component decomposition of treatment response in aggregated N-of-1 trials: pharmacological, expectation-driven, and natural-history components

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30 *This document is a structured, section-by-section summary of the key points argued in the*
31 *companion manuscript, “Three-component decomposition of treatment response in aggregated*
32 *N-of-1 trials: pharmacological, expectation-driven, and natural-history components.” It is pro-*
33 *vided as a supplementary reading aid and does not stand on its own; all notation, derivations,*
34 *simulation detail, and citations are in the main manuscript.*

35

36 1 Introduction

37 **Partitioning treatment response into pharmacological (BR), placebo-belief (PB),**
38 **and natural-history (TV) channels has biological precedent and clinical appeal;**
39 **this paper rigorously evaluates when the partition is worth making.**

- 40 • Precedent runs from pharmacometric disease-progression models [holford2015;
41 @gomeni2009] through Kaptchuk’s empirical decomposition of the placebo response
42 itself [kaptchuk2008components] to neuroimaging dissociations of drug from ex-
43 pectancy effects [wager2013nps; zunhammer2018nps]; each motivates the partition
44 on independent biological grounds.
- 45 • Clinical appeal is direct: BR is what persists after expectancy decays; PB drives appar-
46 ent response in high-placebo-responsiveness populations and misleads precision-medicine
47 stratification when mistaken for pharmacological prediction; TV is what would have
48 changed without any treatment at all.
- 49 • Partitioning costs degrees of freedom, requires specific trial-design contrasts for identifi-
50 ability, and adds analytic complexity; whether the cost is warranted is an empirical and
51 design-dependent question this paper answers.

52 **The optimal choice is conditional: decompose when the candidate biomarker may**
53 **correlate with PB or TV ; use the one-component analysis when domain knowledge**
54 **supports orthogonality.**

- 55 • Decompose: COMT val158met genotype as a biomarker in IBS or PTSD trials – COMT
56 regulates dopaminergic expectancy processing and is a documented predictor of placebo-
57 response magnitude [hall2012comt; wang2022comt], making it a PB moderator
58 whose one-component biomarker-treatment slope conflates pharmacological and belief-
59 driven prediction. Baseline symptom severity is similarly confounded: it predicts phar-
60 macological response but also couples to regression-to-mean TV [hrobjartsson2001;
61 @hrobjartsson2010] and heightened treatment expectancy [kirsch2008; locher2015].
- 62 • One-component sufficient: CYP metabolizer phenotype as a pharmacokinetic biomarker
63 – enzymatic drug metabolism cannot plausibly couple to belief state or untreated disease
64 trajectory, so the one-component pharmacological slope is already the target estimand.
65 ADRB2 Arg16Gly in asthma bronchodilator trials provides an analogous case: the vari-
66 ant affects receptor downregulation via a direct pharmacological mechanism, lung func-

tion is objectively measured by spirometry, and no expectancy pathway links ADRB2 genotype to PB .

- In the PTSD/prazosin context of this paper: baseline PCL-5 severity belongs in the decompose column (a psychological measure with documented coupling to both PB and TV); an adrenergic receptor variant affecting prazosin binding affinity belongs in the one-component-sufficient column (pharmacological mechanism, no PB coupling).

An algebraic identity establishes the conditional rule exactly; Study A confirms the orthogonal arm; Study B (pre-registered, pilot complete) addresses the coupling arm; together they bound the inferential cost of the choice.

- The identity (section 6) is exact: bias in the one-component slope = $w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV}$; it is zero if and only if the biomarker is orthogonal to PB and TV , regardless of the magnitude of those components.
- Study A (section 7.3; 1,000 replicates per alternative cell, 5,000 per null) confirms the orthogonal arm: mean absolute bias in the one-component $\hat{\beta}_{bm:D_{bc}}$ is 0.016 against a true value of -2.25 at $N = 70$, with no trend across $m_{PB} \in \{0, 1, 3, 6, 10\}$ or $m_{TV} \in \{-1, 0, 1, 2\}$.
- Study B (section 7.4; 100-replicate pilot reported here; production run pre-registered) addresses the coupling arm: pilot data show bias scaling with $c_{bm,PB}$, consistent with the algebraic prediction and motivating the production run.

2 Notation and the formal model

This section fixes compact notation and states the formal model; a fuller worked-example treatment is in the companion pedagogical document.

- Definitions are precise enough to support the identifiability and analysis-side arguments in later sections.
- The presentation is kept compact; extended derivations are deferred to the companion document.
- The companion pedagogical document provides numerical worked examples at each formal step.

where Y_{it} is the observed symptom score, BL_i is participant i 's baseline score, the three components BR , PB , TV are non-negative reductions in symptom severity attributable to the three causal mechanisms identified in section 1, and ε_{it} is observation-level noise. The sign convention is that an increase in any component reduces Y ; all three components are zero at $t = 0$ by construction (no exposure yet to drug, no belief activation, no elapsed natural history) and grow over the duration of the trial.

Each component is driven by its own time-on-state variable, making the components structurally distinct even when they overlap in calendar time.

- BR_{it} accumulates with time-on-drug; PB_{it} with time-on-belief scaled by expectancy $\eta(\text{phase})$; TV_{it} with calendar time from trial entry.

105 • The distinct time indices are the structural source of identifiability from phase-contrast
106 data.

107 • All components are zero at $t = 0$ by construction and grow over the trial duration.

108 rescaled so that $C(0) = 0$ and $C(t) \rightarrow m_C(1 - e^{-d_C t})$ as $t \rightarrow \infty$, for $C \in \{BR, PB, TV\}$. The
109 three Gompertz parameters (m_C, d_C, r_C) are participant-specific random effects drawn from
110 population distributions, and the participant-level heterogeneity in these distributions is the
111 principal source of between-participant response variability that the framework is designed to
112 characterize.

113 **The residual term has AR(1) structure and the three components are permitted**
114 **to co-vary through cross-component covariances.**

115 • Within-participant residuals follow $\text{Cov}(\varepsilon_{it}, \varepsilon_{i,t+h}) = \sigma_\varepsilon^2 \rho^{|h|}$.

116 • Within-time and across-time cross-component covariances permit participant-level het-
117 erogeneity in one component to co-vary with the others.

118 • This structure accommodates biologically plausible correlations, such as high-*BR* pa-
119 tients also tending toward high-*PB*.

120 3 The three components

121 **This section presents each component with its biological motivation, Gompertz**
122 **parameterization, and identifiability conditions to establish the model as causally**
123 **interpretable.**

124 • The three components are presented in turn: *BR* (biological response), *PB* (placebo-
125 belief), *TV* (natural history).

126 • Each component is grounded in independent biological literature before its parametric
127 form is stated.

128 • The section's purpose is to demonstrate that the model sum reflects causal structure,
129 not mathematical convenience.

130 3.1 Biological response (BR)

131 ***BR* is the pharmacological target of regulatory approval and biomarker stratifi-**
132 **cation, independent of belief or natural history.**

133 • *BR* arises from chemical action on the body and is zero for inert pills regardless of
134 patient expectation.

135 • It is the estimand that approval decisions, dose selection, and biomarker-guided strati-
136 fication require.

137 • Separating *BR* from *PB* and *TV* is the central precision-medicine problem the frame-
138 work addresses.

139 **Pharmacokinetic/pharmacodynamic (PK/PD) considerations produce a saturat-**
140 **ing *BR* trajectory with a slow initial rise, a peak-growth phase, and a sharp drop**

141 **to zero on discontinuation.**

- 142 • Plasma steady state is reached within a few half-lives; pharmacodynamic integration
143 (e.g. receptor adaptation) adds additional lag.
- 144 • For prazosin-PTSD, Raskind et al. (2013) calibrate the onset to several weeks of sus-
145 tained exposure.
- 146 • The saturating shape motivates the Gompertz over linear or simple exponential alter-
147 natives.

148 **The three Gompertz parameters (m_{BR}, r_{BR}, d_{BR}) capture ceiling, onset rate, and**
149 **climb shape respectively, calibrated to prazosin-PTSD reference values.**

- 150 • m_{BR} is the biological ceiling (maximum reduction at infinite exposure); r_{BR} the onset
151 rate; d_{BR} the displacement shaping the early climb.
- 152 • Prazosin-PTSD calibration yields $m_{BR} \approx 11$, $r_{BR} \approx 0.42/\text{week}$, $d_{BR} \approx 5$.
- 153 • These are participant-specific random effects whose population distributions encode
154 between-participant response heterogeneity.

155 **The Gompertz is preferred over linear and simple-exponential alternatives be-**
156 **cause it captures the asymmetric slow-start saturation profile of chronic-condition**
157 **pharmacodynamics.**

- 158 • Linear $BR = \beta t$ is biologically implausible past the first few weeks due to unbounded
159 growth.
- 160 • Exponential $BR = m(1 - e^{-rt})$ lacks the slow-start phase from receptor adaptation and
161 gene-expression changes.
- 162 • The Gompertz interpolates between both limits and its three-parameter form avoids
163 overfitting; sensitivity analysis is reported separately in 07-gompertz-evaluation.

164 **BR is identified by the blinded-discontinuation contrast, where drug removal col-**
165 **lapses BR to zero while PB remains intact.**

- 166 • Within-participant on-drug versus off-drug contrasts are the primary identification mech-
167 anism.
- 168 • The blinded-discontinuation phase provides the cleanest contrast: BR drops, PB is
169 preserved.
- 170 • This contrast does not require the analyst to model PB or TV explicitly to recover BR .

171 3.2 Placebo-belief response (PB)

172 **PB is the strict placebo response: improvement that persists when the active**
173 **drug is silently replaced, provided the patient's belief is unchanged.**

- 174 • PB is belief-dependent, not drug-dependent; it would survive a silent switch to placebo
175 under sustained belief.
- 176 • It is distinct from BR in causal origin: BR requires chemical action; PB requires only
177 belief.

178 • The definition is strict and operational, not vague or colloquial.

179 ***PB* reflects neurobiologically real mechanisms triggered by belief, not pharma-**
180 **cology; neuroimaging and genetic evidence confirm it is a structured biological**
181 **phenomenon.**

182 • Belief-driven endogenous opioid/dopamine release, HPA-axis modulation, and top-down
183 attentional shifts are all documented mechanisms.

184 • Wager et al. neurologic pain signature dissociates pharmacological from expectancy con-
185 tributions via fMRI.

186 • COMT genotype predicts placebo responsiveness, confirming biological structure rather
187 than measurement artifact.

188 ***PB* is identifiable from *BR* only when belief and treatment allocation differ, as in**
189 **the blinded-discontinuation phase.**

190 • The blinded-discontinuation phase is the canonical contrast: patients cannot tell whether
191 they remain on active drug.

192 • Belief drops to a partial level ($\eta \approx 0.5$) because the patient's expectation is hedged, not
193 absent.

194 • Designs without a phase that decouples belief from treatment cannot separately identify
195 *PB* from *BR*.

196 **The $\eta = 0.5$ blinded-phase expectancy is a modeling assumption operationalising**
197 **response expectancy, not a measured quantity, and requires sensitivity analysis.**

198 • The construct is response expectancy in the sense of Kirsch (1985); to our knowledge
199 the residual expectancy under blinded discontinuation has not been measured directly.

200 • Open-label placebo trials confirm clinically meaningful *PB*-only responses even under
201 explicit disclosure, bounding the residual response away from zero.

202 • Sensitivity analyses varying η are mandatory before clinical conclusions depend on this
203 single parameter.

204 **Kaptchuk et al. (2008) provide the structural precedent; the BR-PB-TV frame-**
205 **work extends the same decomposition logic to the full treatment response.**

206 • Kaptchuk decomposed the placebo response itself into observation, ritual, and relation-
207 ship components empirically.

208 • The present framework applies analogous decomposition logic at the level of drug,
209 placebo, and natural history.

210 • Both frameworks reject the 'expectancy lump' model in favor of identifiable additive
211 parts.

212 ***PB* variance scales with η , not just its mean; ignoring this heteroscedastic struc-**
213 **ture miscalibrates standard errors and downstream biomarker tests.**

214 • Phase-dependent variance follows $\text{SD}(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$.

- Open-label phases have high between-participant PB variance; blinded phases have low variance.
- Ignoring heteroscedasticity deflates standard errors in open-label and inflates them in blinded phases, with consequences for CI coverage and biomarker test calibration.

3.2.1 Additivity, subadditivity, and the BR-PB interaction

Additivity is a modeling assumption; published literature documents subadditivity between pharmacological and belief effects in some contexts; the phase-augmented analysis partly absorbs it as a side benefit.

- The generalized model adds $\gamma \cdot BR_{it} \cdot PB_{it}$; $\gamma < 0$ encodes subadditivity, $\gamma = 0$ recovers the additive form.
- Apparent subadditivity can arise from latent-class structure with class-correlated BR and PB ; distinguishing true interaction from confounding requires a class-aware analysis or the phase contrasts of section 4.
- Simulation quantifying bias scaling with γ is deferred to the production program (section 7).

3.3 Time-variant natural-history component (TV)

TV is the untreated counterfactual trajectory; it is participant-specific, structured, and can be positive, negative, or near-zero depending on disease class.

- TV is the symptom change that would have occurred without treatment, integrated over the trial duration.
- It varies in sign across disease classes and across patients within a class.
- It is independent of treatment status and belief, distinguishing it structurally from PB .

TV is a structured participant-specific signal, not noise; its distinguishing feature from PB is independence from treatment status and belief.

- Unlike ε_{it} , TV is participant-level (slow trajectory) and requires explicit modeling, not absorption into noise.
- Off-drug baselines and follow-up windows are the primary data sources for estimating TV .
- TV is modulated by neither treatment assignment nor patient belief, whereas PB is modulated by both.

3.3.1 Natural history: existence, testability, and functional form

TV sign varies by disease class; the zero- TV assumption is rarely safe in chronic trials; a linear week covariate is insufficient because it enforces population-mean linearity and cannot separate TV from BR without the design contrasts of section 4.

- Disease class determines TV sign: positive for acute recovery, near-zero for stable chronic, negative for progressive conditions; selection-driven enrollment adds a positive TV component regardless of class due to regression to the mean [hrobjartsson2001; hrobjartsson2010].
- Zero- TV is safe only when all four conditions hold: stable disease, short trial, non-severity-peaked recruitment, no ancillary care; in most chronic trials at least one fails.
- A **week** covariate enforces linearity, ignores participant-level heterogeneity, and conflates BR with TV ; the Gompertz form and the design contrasts of section 4 supply identifiability.

4 Identifiability and trial-design contrasts

Identifiability is supplied by trial design; each phase of the hybrid design contributes a different combination of components to the observed trajectory.

- The decomposition is mathematically specified but requires trial-design contrasts to be statistically identified.
- This section characterizes the hybrid open-label-plus-blinded-discontinuation design as the primary identifiability vehicle.
- Each of the three phases supplies a distinct mixture of BR , PB , and TV .

Three phase-by-allocation contrasts isolate PB , BR , and TV respectively; the LME model combines them but the inference is driven by design.

- Open-label versus blinded contrast yields approximately $0.5 \cdot PB$, isolating the belief component conditional on a known blinded-phase η .
- Blinded on-drug versus blinded placebo yields BR once drug carryover has elapsed (since PB and TV match across arms within the blinded phase).
- Off-drug phases yield TV plus residual off-drug PB , with no active BR contribution beyond carryover.

Designs lacking specific phases lose identifiability of the corresponding component; the decomposition machinery is only as useful as the design allows.

- A pure open-label design cannot separate BR from PB because both are present at full strength throughout.
- A pure blinded design cannot estimate $\eta = 1$ PB because no phase achieves full-strength belief.
- Trial designers should verify each component has at least one phase in which it is uniquely present before committing to the design.

Design-contrast identifiability is necessary but not sufficient: the three Gompertz trajectories are collinear, so joint estimation is credible only under specific design and sample-size conditions.

- The components share an onset shape and overlap in calendar time, so their parameters

- 287 can be near-collinear even when the identifying contrasts are present.
- 288 • Joint recovery requires a phase in which each component varies while the others are held
 - 289 fixed, and an off-drug window long enough for *TV* to separate from a decaying *BR*.
 - 290 • The recoverable region of the design-and-sample-size space is mapped by the pre-
 - 291 registered identifiability diagnostics, not asserted.

292 5 Analysis-side methodology

293 **Four constraints make a component-matched analysis impractical at trial-relevant**
 294 **sample sizes; a phase-augmented LME captures the important part of the gap at**
 295 **minimal cost.**

- 296 • The question of whether to mirror the data-generating process (DGP) in the analysis
- 297 model is reasonable; the section addresses it directly.
- 298 • The four constraints are identifiability, degrees of freedom, convergence fragility, and
- 299 marginal estimand robustness.
- 300 • A phase-by-treatment indicator is the recommended modest extension that captures
- 301 BR-versus-PB sensitivity without the full parametric cost.

302 5.1 Four reasons not to match the DGP

303 **The first constraint is identifiability: forcing a three-component model on a de-**
 304 **sign with insufficient contrasts produces unidentified parameters, not component**
 305 **estimates.**

- 306 • A pure open-label trial cannot distinguish *BR* from *PB*; both rise from zero together
- 307 on treatment initiation.
- 308 • Fitting three components to a one-contrast design produces implausibly tight standard
- 309 errors that misrepresent inferential precision.
- 310 • The analyst must audit whether the design supplies the necessary contrasts before spec-
- 311 ifying a component-matched model.

312 **The second constraint is degrees of freedom: a component-matched analysis adds**
 313 **approximately a dozen non-linear parameters, inflating moderation estimand vari-**
 314 **ance at $N \in [30, 150]$.**

- 315 • Gompertz *BR* (3), *PB* (3 + expectancy multiplier), and *TV* (3) each require participant-
- 316 specific random effects on the maxima.
- 317 • At trial-relevant N , the variance of $\hat{\beta}_{bm:D}$ is dominated by participant and timepoint
- 318 counts.
- 319 • Additional parametric structure typically inflates moderation estimand variance more
- 320 than it reduces bias.

321 **The third constraint is convergence fragility: component-matched nonlinear**
 322 **mixed models converge below 90% at trial-relevant N , contaminating aggregated**

analyses.

- Simultaneous Gompertz BR , PB , and TV fitting requires `nlme::nlme`, Bayesian hierarchical, or structural-equation approximations.
- Each is more sensitive to starting values, prone to local optima, and likely to fail on individual trial datasets than the LME analogue.
- Non-converging cells at below-90% convergence rates contaminate power and bias estimates in simulation studies.

The fourth constraint is estimand robustness: the `bm:Dbc` marginal interaction is the right estimand for the biomarker-validation question regardless of the BR-PB split.

- Regulators want to know whether the biomarker predicts treatment response under trial conditions, not a mechanistically decomposed prediction.
- The marginal `bm:Dbc` interaction captures this question whether the moderation is biological or expectation-mediated.
- Component decomposition adds mechanistic precision but is not required for the core biomarker-validation decision.

The inferential target is stated as an estimand: the population biomarker-treatment interaction with explicit intercurrent-event handling, while the pharmacological-component target is more ambitious and inherits the heterogeneity-identifiability caveat.

- The $bm:D_{bc}$ estimand is defined by its population, treatment contrast, endpoint, summary, and intercurrent-event strategy in the ICH E9(R1) sense.
- Dropout is handled by a treatment-policy strategy and residual carryover by the modeled exponential decay.
- Recovering a participant-level pharmacological interaction inherits Senn's caveat that true heterogeneity is separable from within-person variation only under repeated phase contrasts.

5.2 A small extension that usually pays off: phase indicators

The phase-augmented LME uses `Dbc:phase` and `bm:Dbc:phase` to test whether drug and biomarker-by-drug effects attenuate under blinding.

- `phase` absorbs systematic differences in PB expectancy across trial phases.
- `Dbc:phase` detects whether the drug effect shrinks under blinding, a signature of PB -mediation.
- `bm:Dbc:phase` detects whether the biomarker-treatment interaction shrinks under blinding, a signature that it is partly biomarker-by- PB .

The phase-augmented extension adds at most four parameters yet captures the practically relevant belief-attenuation diagnostic without requiring the Gompertz parametric form.

- Three to four parameters is well within budget at any realistic trial sample size.
- The extension does not separately estimate *BR* and *PB* components but identifies the change in apparent drug effect when belief is hedged.
- For most analyses this is the practically relevant question and is answerable without committing to the Gompertz form.

5.3 When to fit the full decomposition

The full decomposition merits the effort only when N is in the hundreds, the design is fully phase-balanced, and the scientific question requires the BR-PB mechanistic split.

- Feasibility conditions: several-hundred N , dense timepoints in every phase, mechanistic BR-PB split required, and informative Bayesian priors available.
- In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or `brms` can return participant-specific component estimates.
- Outside that regime the phase-augmented LME is the better tool.

The DGP-analysis asymmetry is deliberate: simulation uses the rich DGP to characterize how a simpler analysis behaves; the analysis need not mirror the DGP to produce useful inference.

- Simulation controls the DGP and can specify any structure; inference must pay the identifiability cost of every parameter.
- The DGP is a characterization tool; the analysis is a finite-data inference tool; they have different optimality criteria.
- DGP-matching is not required and is often counterproductive at trial-relevant sample sizes.

The exponential-decay carryover indicator is a deliberately parsimonious special case of richer carryover models, and the simulation should benchmark it against a distributed-lag alternative.

- The continuous indicator D_{bc} decays as $\exp(-\lambda t_{sd})$, a one-parameter form tied to a pharmacokinetic half-life.
- Distributed-lag models, the average-period-treatment-effect estimand, and G-estimation relax the single-rate assumption at additional parametric cost.
- The exponential restriction is an assumption, not a finding, and the pre-registered simulation should include a distributed-lag arm to quantify its cost.

5.4 What the one-component analysis estimates

The one-component treatment-effect estimate has a probability limit equal to the pharmacological component plus belief and natural-history contributions, so its bias does not vanish as N grows.

- Writing the reduction as $BR+PB+TV-\varepsilon$ and fitting one drug indicator gives $\text{plim } \hat{\beta}_D = m_{BR} + b_{PB} + b_{TV}$.
- b_{PB} and b_{TV} are the belief and natural-history terms that load onto the drug indicator under the chosen contrast; both are generally non-negative for an improving outcome.
- Because the displacement is in the probability limit, a larger one-component trial converges on the same biased value: the failure is identification, not precision.

The biomarker slope from a one-component analysis decomposes additively into component-specific slopes, so a biomarker correlated with PB or TV registers as a predictor even with no pharmacological association.

- By linearity of covariance, $\beta_{bm}^{\text{lumped}} = \beta_{bm}^{BR} + w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV}$.
- A non-zero β_{bm}^{PB} or β_{bm}^{TV} produces a non-zero one-component slope even when the pharmacological slope β_{bm}^{BR} is exactly zero.
- The result is an algebraic identity under the additive decomposition, not a sample-size-dependent tendency.

The bias displaces the estimand rather than inflating variance, so a significance test of the one-component coefficient cannot detect it; the decomposition is what locates it.

- The one-component coefficient is displaced from m_{BR} by $b_{PB} + b_{TV}$ but remains a well-estimated quantity with a valid Wald test of its own nullity.
- Detecting the bias requires comparing $\hat{\beta}_D$ to m_{BR} , a comparison only the decomposition or an isolating phase contrast supplies.
- This is why a rejection-rate summary is the wrong instrument, and accounts for the flat rejection rates across the placebo-strength axis in the section 7.1 pilot.

5.5 A numerical illustration

Figure 1 illustrates the diagnostic contrast: two participants with identical open-label responses but opposite BR - PB profiles become distinguishable only in the blinded-discontinuation phase.

- Participant A ($m_{BR} = 4$, $m_{PB} = 6$) and Participant B ($m_{BR} = 8$, $m_{PB} = 1$) both show approximately ten points of open-label improvement, indistinguishable to a one-component analysis that estimates drug at 7.75 points.
- In the blinded-discontinuation phase A retains four points (PB persists at $\eta = 0.5$) while B collapses to 1.5 points (drug cleared, minimal PB): a seven-point diagnostic contrast.
- The three-component decomposition recovers $\hat{m}_{BR}^A = 4$, $\hat{m}_{BR}^B = 8$, $\hat{m}_{PB}^A = 6$, $\hat{m}_{PB}^B = 1$ – six clinically actionable quantities versus one biased average; predictive-biomarker validation requires these.

6 Simulation study

The simulation study follows ADEMP and quantifies bias and identifiability properties of three analysis strategies across a prazosin-PTSD-calibrated parameter grid.

- The Morris (2019) ADEMP framework structures the study: Aims, DGP, Estimands, Methods, Performance measures.
- The pre-registration is at `analysis/scripts/component-decomposition/00-ademp-pre-registration`.
- Three analysis strategies (one-component, phase-augmented, full decomposition) are compared across the parameter grid.

6.1 Pilot validation

A 100-replicate pilot confirmed the analysis pipeline is operational; two design-level findings inform the production specification.

- Convergence was 1.0 across six cells at $N = 35$; per-replicate output is checkpointed at `analysis/data/quick-sim/`.
- At $N = 35$, rank-deficiency dropped the D_{bc} :phase interaction term, requiring larger N and a richer design in the production program.
- The predicted bias manifests in the mean of $\hat{\beta}_{bm:D_{bc}}$, not the Wald-test rejection rate; the production program reports coefficient means against true values.

6.2 Production-simulation specification

The production program follows the ADEMP framework with four deliverables; the realised Study A run in section 7.3 is the first of these.

- The deliverables are interaction-estimate bias, full-decomposition identifiability, Type I error and power, and participant-level signal recovery.
- Sample sizes $N \in \{35, 70, 100, 150\}$ cross a `pb_strength` grid; type I cells target MCSE 0.003 at 5,000 replicates.
- Driver scripts, checkpointed outputs, and the ADEMP pre-registration reside under `analysis/scripts/component-decomposition/`.

6.3 Study A: orthogonal-biomarker arm (1,000 replicates per alternative cell)

Study A is the orthogonal-biomarker arm of the conditional: with $c_{bm,PB} = 0$, the one-component estimate is essentially unbiased across the entire (m_{PB}, m_{TV}) grid, exactly as the algebraic identity requires.

- The true biomarker-by- BR effect is -2.25 ; mean absolute bias is 0.016 at $N = 35$, 0.016 at $N = 70$, 0.006 at $N = 100$, and 0.006 at $N = 150$, in every case within one MCSE of zero.

- Bias shows no monotone trend across $m_{PB} \in \{0, 1, 3, 6, 10\}$ or $m_{TV} \in \{-1, 0, 1, 2\}$; it behaves as sampling noise around zero.
- This is the predicted consequence of the identity in section 6.1: the DGP couples the biomarker to BR only, so $\beta_{bm}^{PB} = \beta_{bm}^{TV} = 0$ and the one-component slope is unbiased by construction.

The phase-augmented analysis provides no bias reduction and incurs a substantial power penalty, so it is dominated under this uncontaminated DGP.

- Mean power at $N = 35$ is 0.35 (phase-augmented) versus 0.59 (one-component); at $N = 70$, 0.62 versus 0.90; both reach approximately 1.0 by $N = 100$.
- Phase-augmented mean absolute bias is no smaller than one-component at any N (0.028 versus 0.016 at $N = 35$).
- Convergence is approximately 100% for both analyses at every N (minimum 0.998), so the rank-deficiency that degraded the $N = 35$ pilot does not recur here.

Study A confirms the identity but cannot exhibit the biomarker-validation failure; a contaminated-biomarker pilot reported in the next subsection provides the decisive evidence.

- As pre-registered, Study A varies component magnitude with the biomarker fixed to BR , so it probes an axis the identity predicts to be null.
- The decisive axis is $c_{bm,PB}$, the biomarker- PB correlation; this is varied in the contaminated-biomarker pilot below.
- On current evidence the phase-augmented analysis is not recommended for the uncontaminated-biomarker regime; the pilot below evaluates it under contamination.

6.4 Study B pilot: coupling-arm simulation (100 replicates per cell)

A pilot varying the biomarker- PB correlation $c_{bm,PB} \in \{0, 0.15, 0.30, 0.45\}$ confirms that bias in the one-component estimator scales with coupling strength, not component magnitude, and that the phase-augmented analysis provides no protection.

- At $N = 150$, mean $\hat{\beta}_{bm:D_{bc}}$ shifts from -2.25 (true) to -1.74 at $c_{bm,PB} = 0.45$ (bias $+0.51$, 30 MCSEs from zero), confirming the identity's prediction.
- The bias is driven by $c_{bm,PB} \times \sigma_{PB}$, not by the population-mean m_{PB} : cells with $m_{PB} = 0$ and $m_{PB} = 6$ show identical bias at every $c_{bm,PB}$ level, because the individual-level PB variance ($\sigma_{PB} = 2$) provides the contamination channel even when the mean is zero.
- The phase-augmented analysis shows the same bias as the one-component analysis at every contamination level, offering no protection against biomarker- PB coupling.
- This pilot uses 100 replicates per cell; a production run at 1,000 replicates per cell is required to confirm the pattern and assess Type I error under contamination.

The contaminated-biomarker pilot reframes the paper's practical guidance: the risk of biomarker-validation failure arises from biomarker- PB correlation in the

506 **study population, not from the presence of strong placebo response per se.**

- 507 • A trial in a population where the biomarker is uncorrelated with placebo responsiveness
508 can use the one-component estimator without bias, regardless of how large the placebo
509 response is.
- 510 • A trial where the biomarker covaries with optimism, suggestibility, or treatment-seeking
511 history is at material risk of contamination; bias of +0.51 (23% of the true effect) at
512 $c_{bm,PB} = 0.45$ illustrates the practical scale.
- 513 • Neither the one-component nor the phase-augmented analysis removes this bias; full
514 component decomposition is required.

515 **6.5 Study C: recovery under a belief-decoupling design**

516 **Study C shows the contamination is recoverable, but only under a design that de-**
517 **couples drug exposure from belief: a belief-covariate decomposition on a balanced-**
518 **placebo design returns the unbiased pharmacological slope across the feasible**
519 **contamination range, while the one-component estimator does not.**

- 520 • In the multivariate normal (MVN) process the biomarker- BR coupling loads on the drug
521 state D_{bc} and the biomarker- PB coupling scales with the expectancy η ; the decomposi-
522 tion $Y \sim bm + t + D_{bc} + \eta + bm:D_{bc} + bm:\eta$ isolates β_{bm}^{BR} on $bm:D_{bc}$ only if D_{bc} and η
523 vary independently.
- 524 • On the standard hybrid design they are collinear (analysis-stage correlation $\approx +0.6$),
525 and the belief-covariate, blinded-stratum, and Gompertz-basis decompositions all fail
526 there, no better than the one-component analysis.
- 527 • A balanced-placebo design adds a covert on-drug phase ($\eta = 0$) and an open-placebo
528 phase ($\eta = 1$), decoupling the contrasts (≈ -0.66) and enabling recovery.

529 **At 1,000 replicates per cell the decomposition is unbiased across the contamina-**
530 **tion range while the one-component estimator degrades monotonically; recovery**
531 **is conditional on the design and bounded by positive-definiteness.**

- 532 • At $N = 150$ the decomposition bias of $\hat{\beta}_{bm:D_{bc}}$ stays within one MCSE of zero across
533 $c_{bm,PB} \in \{0, 0.1, 0.2, 0.3\}$ ($-0.022, +0.008, +0.021, -0.026$; MCSE ≈ 0.02), while the
534 one-component bias rises to $+0.16, +0.34, +0.48$.
- 535 • Interval coverage is near-nominal for the decomposition (0.94–0.96) and falls to 0.86 for
536 the one-component analysis at the highest contamination; convergence is 100 percent.
- 537 • The one-component power advantage is confined to low contamination: it is $+0.19$
538 (power 0.85 versus 0.66) at $c_{bm,PB} = 0$, $N = 70$, erodes monotonically as contam-
539 ination grows, and reverses by $c_{bm,PB} = 0.3$ (0.64 versus 0.67), because the biased
540 one-component estimate is dragged toward the null.
- 541 • The grid is capped at $c_{bm,PB} = 0.30$ because beyond approximately 0.45 the implied
542 covariance is non-positive-definite and the contaminated DGP is no longer faithfully
543 representable.

7 Application to predictive-biomarker validation

The precision-medicine question – whether a biomarker predicts pharmacological response – is answered by the one-component total only when the biomarker is uncorrelated with PB and TV ; otherwise it requires decomposing the response.

- The question is whether the biomarker predicts the BR component, not the one-component total.
- The PATH statement formalizes this as the target estimand for predictive biomarker validation.
- The BR-PB-TV decomposition is the mechanism that separates the pharmacological signal when biomarker-component coupling cannot be ruled out.

A one-component-total biomarker regression conflates pharmacological and placebo-belief prediction, producing a stratification biomarker that mis-identifies responders.

- $\hat{\beta}_{bm}$ from the one-component regression contains both pharmacological and placebo-belief prediction in unknown proportions.
- If PB covaries with the biomarker distribution, flagged 'high responders' include high- PB responders who will not benefit pharmacologically.
- Prospective application of such a biomarker would systematically mis-stratify patients for prescription decisions.

A valid pharmacological predictor requires a non-zero $\hat{\beta}_{bm}^{BR}$; significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ indicate a psychological or prognostic biomarker, not a pharmacological one.

- The sole criterion for pharmacological prediction is a meaningfully non-zero BR -component regression coefficient.
- A biomarker correlating only with PB selects placebo responders, not drug responders.
- A biomarker correlating only with TV is prognostic but not predictive in the pharmacological sense.

When biomarker-component coupling cannot be ruled out, the decomposition is the mechanism by which the precision-medicine question is posed; when it can, the one-component analysis already answers it.

- Where the biomarker may correlate with PB or TV , separating BR from them is what prevents validation from conflating pharmacological and non-pharmacological prediction.
- The phase-augmented `bm:Dbc:phase` interaction is a small- N proxy, but Study A shows it buys nothing under an uncontaminated biomarker, so it is warranted only when contamination is plausible.
- The full decomposition recovers component-specific regressions explicitly when sample size permits.

8 Discussion

The three-component framework reframes treatment response as a structured sum of causally distinct contributions, and the present work maps when acting on that decomposition actually changes inference.

- The framework specifies which trial-design contrasts identify which components.
- Acting on the decomposition is valuable conditionally: for attribution estimands, and for biomarkers that may correlate with PB or TV , but not for a biomarker-treatment interaction with a biomarker orthogonal to them.
- The target estimand for biomarker validation is the biomarker-by- BR association; whether a one-component analysis already recovers it depends on the biomarker's coupling to PB and TV .

Four open questions bound the framework's current scope: the simulation results are pending, the η multipliers require sensitivity analysis, non-monotone natural history is not yet handled, and participant-level recovery inherits the identifiability caveats of individual treatment effects.

- Quantitative bias-versus-power trade-offs are conjectures until the section 7 simulation is run.
- The η (phase) expectancy-modulation parameters are modeling assumptions requiring explicit sensitivity analysis.
- Cyclical or biphasic TV trajectories exceed the Gompertz form and require flexible or covariate-augmented specifications.
- Participant-level component recovery rests on separating participant-by-component signal from participant-by-period and participant-by-phase variation, the identifiability caveat Senn raises for individual treatment effects.

Two companion documents extend the manuscript: a pedagogical narrative and a Gompertz-family sensitivity evaluation.

- The pedagogy document at `docs/24-component-decomposition-pedagogy.md` provides worked examples for first-time readers.
- The Gompertz evaluation at `analysis/report/07-gompertz-evaluation/` quantifies trajectory-specification sensitivity.
- Together they address the intuitive and parametric dimensions of the framework that the present manuscript treats briefly.

The components exist in any longitudinal treatment response, but participant-level identifiability requires a within-subject belief-by-exposure contrast that not every placebo-controlled design supplies.

- The three-component structure is a property of the data-generating process of any repeated-measures treatment study; its identifiability is design-specific.
- Parallel-group placebo control identifies the population-mean PB but not a participant-

level PB , and supplies no within-subject belief contrast; only designs with such a contrast (blinded or open-hidden discontinuation, randomized withdrawal, crossover, hybrid N-of-1) identify the participant-level decomposition the biomarker question needs.

- The broad claim is therefore scoped to designs supplying a within-subject belief-by-exposure contrast, not to placebo-controlled longitudinal trials in general.

9 Conclusions

Treatment response in aggregated N-of-1 trials is the sum of three causally distinct components, and the bias from conflating them is governed by the estimand and the biomarker's coupling to the components, not by their raw magnitude.

- The three components are BR (pharmacological), PB (placebo-belief), and TV (natural history).
- Each has its own biological mechanism, identifiability conditions, and clinical interpretation.
- The one-component pre-post treatment effect is biased by the PB and TV contributions; the biomarker-treatment slope is biased only by the biomarker's covariance with PB and TV , as Study A confirms.

What matters is the design contrast, not the parametric model: the trial must supply the drug-on/off and belief contrasts, but adding parametric component terms is not free and is not always worth it.

- Unique component identification requires phase contrasts built into the trial design; the one-component analysis already exploits the drug-on/off contrast.
- Under an uncontaminated biomarker, the phase-augmented analysis adds no bias reduction and loses power, so it is not recommended as a default.
- The phase-augmented and full component-matched analyses are warranted for attribution estimands and for biomarkers that may correlate with PB or TV , where the extra structure does work.

For predictive-biomarker validation the decomposition matters precisely when the candidate biomarker may correlate with PB or TV ; this is the case the decisive simulation still has to probe.

- A biomarker that predicts the one-component total is a pharmacological predictor only if it is uncorrelated with PB and TV ; otherwise the one-component slope is contaminated.
- Regulatory and prescribing decisions therefore require separating BR from PB and TV whenever biomarker-component coupling cannot be ruled out.
- Study A established the uncontaminated baseline; the contaminated-biomarker cell is the planned empirical test of the failure mode.

654 10 Reproducibility

655 The manuscript is rendered inside a Docker container with locked package ver-
656 sions; the full software environment is pinned by Dockerfile and renv.lock.

- 657 • R 4.4.x in the zzcollab Docker container provides the execution environment.
- 658 • Simulation packages are nlme, parallel, data.table, furrr, and purrr.
- 659 • The manuscript build uses rmarkdown and bookdown.

660 Reproducibility is ensured by a master seed of 20260507 with per-replicate seeds
661 derived by index addition.

- 662 • `set.seed(20260507)` is called at the top of each replicate loop.
- 663 • `parallel::mc.set.seed` propagates seeds inside `mclapply`.
- 664 • Per-replicate seeds are `master_seed + rep_idx`.

665 All simulation outputs, driver scripts, pre-registration, and manuscript source
666 files are versioned at documented paths within the repository.

- 667 • Per-replicate outputs and Morris summaries are at `analysis/data/quick-sim/`.
- 668 • Driver scripts and the ADEMP pre-registration are at `analysis/scripts/component-decomposition/`.
- 669 • Manuscript source, bibliography, and CSL file are at `analysis/report/06-component-decomposition/`.